

# OPTIONS PRICING 101

---

A COMPLETE GUIDE TO OPTIONS DATA, GREEKS,  
IV SURFACE, DEALER HEDGING & CONFLUENCE ANALYSIS

This reference covers every layer of professional options analysis — from basic contract mechanics and the options chain through the full Greek risk framework, implied volatility surface interpretation, dealer gamma/delta exposure (GEX/DEX), and how to read data confluence without over- or under-weighting any single signal.

01 Option Fundamentals

02 Options Chain Anatomy

03 The Greeks in Full

04 IV Surface & Skew

05 GEX · DEX · Dealer Hedging

06 Options Pricing & BSM

07 Confluence Framework

08 Common Structures

09 Key Principles

01

OPTION FUNDAMENTALS

CONTRACTS · RIGHTS · OBLIGATIONS

An **option** is a derivative contract giving the buyer the **right, but not the obligation**, to buy or sell an underlying asset at a predetermined **strike price** on or before a specific **expiration date**. The buyer pays a **premium** for this right; the seller (writer) collects the premium and takes on the obligation to fulfill the contract if exercised. Options derive their value from four core inputs: underlying price, time remaining, implied volatility, and the risk-free rate.

|                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>CALL OPTION</div> <div>Grants the holder the right to <b>BUY</b> the underlying at the strike price. Call buyers are <i>long delta</i> — they profit as price rises. Call sellers are <i>short delta</i> with theoretically unlimited upside risk. Calls gain value when price rises, IV expands, or when time decay slows near an ATM strike.</div> <div>Long Call = Long Delta · Bullish Bias · Defined Risk (Buyer)</div> | <div>PUT OPTION</div> <div>Grants the holder the right to <b>SELL</b> the underlying at the strike price. Put buyers are <i>short delta</i> — they profit as price falls. Put sellers take on significant downside exposure. Puts gain value when price falls, IV expands, or when skew steepens during a flight-to-safety environment.</div> <div>Long Put = Short Delta · Bearish Bias · Defined Risk (Buyer)</div> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

MONEYNESS — HOW STRIKES RELATE TO SPOT PRICE

| CONDITION             | CALL STATUS        | PUT STATUS         | INTRINSIC VALUE  | EXTRINSIC VALUE          |
|-----------------------|--------------------|--------------------|------------------|--------------------------|
| Spot > Strike (S > K) | IN-THE-MONEY (ITM) | OUT-OF-MONEY (OTM) | S – K (positive) | Premium above intrinsic  |
| Spot ≈ Strike (S ≈ K) | AT-THE-MONEY (ATM) | AT-THE-MONEY (ATM) | ≈ \$0            | All premium is extrinsic |
| Spot < Strike (S < K) | OUT-OF-MONEY (OTM) | IN-THE-MONEY (ITM) | K – S (positive) | Premium above intrinsic  |

02

OPTIONS CHAIN ANATOMY

READING THE BOARD · CALLS LEFT · PUTS RIGHT

The **options chain** is the central display of all available contracts for a given underlying and expiry. Calls appear on the left, puts on the right, with strikes running down the center. **In-the-money rows are shaded** — green for calls (spot above strike), red for puts (spot below strike). The ATM row carries the highest gamma and is the most actively traded. Reading the full chain reveals OI concentrations, skew, and where dealer hedging forces are most concentrated.

SPX ILLUSTRATIVE CHAIN · SPOT: 5,482 · WEEKLY EXPIRY

| DELTA | IV % | OI     | BID   | ASK   | STRIKE | BID    | ASK    | OI     | IV % | DELTA |         |
|-------|------|--------|-------|-------|--------|--------|--------|--------|------|-------|---------|
| 0.76  | 14.8 | 2,810  | 82.10 | 82.50 | 5400   | 0.55   | 0.65   | 12,400 | 17.6 | -0.12 |         |
| 0.68  | 15.1 | 3,900  | 52.60 | 53.00 | 5430   | 1.80   | 2.00   | 19,200 | 16.9 | -0.22 |         |
| 0.61  | 15.4 | 7,650  | 34.80 | 35.20 | 5450   | 3.40   | 3.70   | 28,900 | 16.5 | -0.31 |         |
| 0.50  | 15.8 | 22,400 | 18.40 | 18.70 | 5480 ♦ | 17.90  | 18.20  | 31,200 | 15.9 | -0.50 | AT<br>M |
| 0.38  | 16.2 | 18,100 | 9.20  | 9.50  | 5510   | 32.10  | 32.50  | 14,800 | 16.8 | -0.64 |         |
| 0.26  | 16.9 | 31,500 | 3.80  | 4.10  | 5540   | 57.20  | 57.70  | 9,400  | 17.4 | -0.76 |         |
| 0.08  | 19.2 | 62,800 | 0.30  | 0.45  | 5620   | 118.50 | 119.20 | 4,100  | 19.9 | -0.93 |         |

OI (OPEN INTEREST)

Total outstanding contracts at a strike. Large OI concentrations act as gravitational zones near expiry — dealers with large gamma books at those strikes must actively hedge, which can create "pin" behavior as price approaches the strike.

BID / ASK SPREAD

Represents market maker liquidity. Tight spreads at ATM confirm active two-way flow. Widening spreads in OTM wings signal lower liquidity and where skew premium is most visible — you pay more to buy protection there.

IV % (COLUMN)

Each strike's implied volatility. Notice that OTM puts carry higher IV than OTM calls (the skew). ATM IV is lowest. Reading IV column by column across strikes tells you the shape of the current volatility surface.

GREEN / RED SHADING

Green rows (left side) = call ITM — spot is above that strike. Red rows (right side) = put ITM — spot is below that strike. The ATM row (yellow) is where gamma is highest and option sensitivity to spot moves is greatest.

03

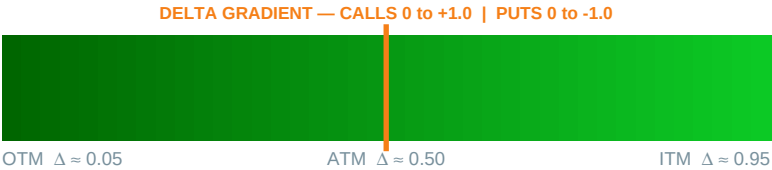
THE OPTION GREEKS

SENSITIVITY MEASURES · RISK DECOMPOSITION

The **Greeks** quantify how an option's price responds to changes in specific market inputs. They do not operate independently — they form an interconnected risk web. Mastering their relationships is the foundation of dealer hedging logic, portfolio construction, and identifying when options positioning creates mechanical price effects in the underlying market.

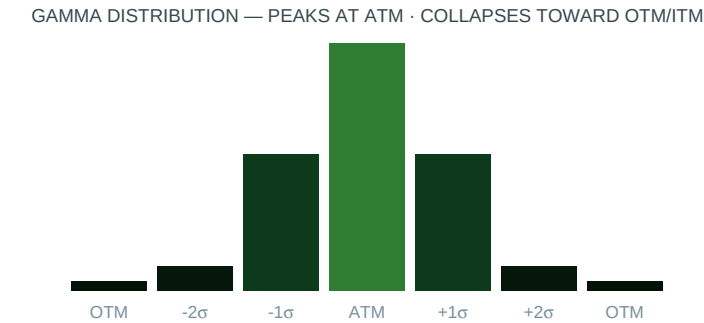
Δ DELTA — DIRECTIONAL SENSITIVITY (FIRST-ORDER)

| DEFINITION                                                                                                               | RANGE                                          | HOW DEALERS USE IT                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rate of change of option price per \$1 move in the underlying. Also approximates the probability the option expires ITM. | 0 to +1.0 (calls) 0 to -1.0 (puts) ATM ≈ ±0.50 | When dealers sell calls, they go short delta and must buy the underlying to stay neutral — this is the core mechanical flow. When they sell puts, they go short delta and must sell the underlying. |



Γ GAMMA — CONVEXITY / RATE OF DELTA CHANGE (SECOND-ORDER)

**Gamma** is the rate of change of delta per \$1 move in the underlying — it measures how quickly your directional exposure shifts. **ATM gamma is highest**, meaning the position's delta changes fastest near the strike. This creates the largest dealer re-hedging activity at ATM strikes. **Positive gamma** (long options) means you benefit from large moves in either direction — your delta grows in the profitable direction. **Negative gamma** (short options) means your losses accelerate as the market moves against you.



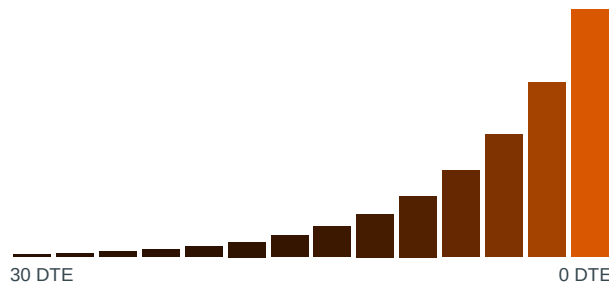
**Formula:**  $\Gamma = \partial^2 V / \partial S^2 = \partial \Delta / \partial S$  · Peaks ATM · collapses exponentially toward OTM/ITM · accelerates near expiry

Θ THETA — TIME DECAY (P&L; EROSION)

**Theta** is the daily erosion of an option's extrinsic value, all else equal. Option **buyers pay theta** — it is the continuous cost of holding optionality. Option **sellers collect theta**, which is their structural edge in a low-volatility environment. The critical relationship: **theta and gamma are always opposite in sign for a long option**. Owning gamma (convexity) means paying theta (time). You cannot have one without the other. Theta accelerates non-linearly in the final 7–14 days before

expiry, which is the engine behind the ODTE and weekly options premium decay cycle.

THETA DECAY ACCELERATION — FINAL 7-14 DAYS = STEEPEST EROSION

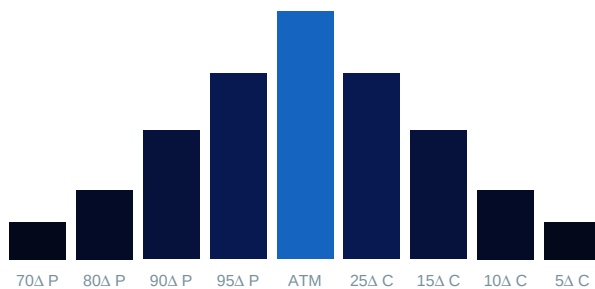


**Formula:**  $\Theta = \partial V / \partial t$  · Always negative for long options · Accelerates sharply near expiry

#### V VEGA — IMPLIED VOLATILITY SENSITIVITY

**Vega** measures how much an option's price changes for every 1% change in implied volatility. Long options are always **long vega** — they benefit from IV expansion. Short options are **short vega** — they suffer when vol rises. **Vega is highest for ATM options and scales with the square root of time to expiry** — a 6-month option has more vega than a 1-week option at the same strike. **IV crush** — the collapse of implied vol after earnings, FOMC, or major events — is the mechanism by which option sellers profit even when direction was correct for buyers.

VEGA BY STRIKE — PEAKS ATM · LONGER DTE = MORE VEGA



**Formula:**  $V = \partial V / \partial \sigma$  · Maximum ATM · Scales with  $\sqrt{\text{DTE}}$  · Primary risk after binary events

**KEY RELATIONSHIP — GAMMA / THETA / VEGA TRIANGLE:** For a long ATM straddle, you are long gamma (benefit from large moves), short theta (pay daily decay), and long vega (benefit from IV expansion). The break-even question is always: will realized volatility exceed implied volatility? If  $RV > IV \rightarrow$  gamma wins. If  $RV < IV \rightarrow$  theta and IV crush win.

04

## IMPLIED VOLATILITY SURFACE

TERM STRUCTURE · SKEW · SMILE · SMIRK

The **IV surface** maps implied volatility across all available strikes (x-axis) and expirations (z-axis). In a perfect Black-Scholes world, IV would be flat — every option on the same underlying would trade at identical vol. In reality, the surface is **tilted, curved, and term-structured**. The shape encodes the market's collective fear, positioning demand, and expectations for tail events across time.

IV SURFACE HEATMAP · SPX ILLUSTRATIVE · RED = HIGH VOL · GREEN = LOW VOL

| EXPIRY | 70Δ P | 80Δ P | 90Δ P | ATM  | 25Δ C | 15Δ C | 5Δ C |
|--------|-------|-------|-------|------|-------|-------|------|
| 0DTE   | 32.1  | 27.4  | 22.1  | 16.2 | 17.1  | 18.4  | 23.8 |
| 1W     | 29.4  | 25.2  | 20.8  | 15.8 | 16.2  | 17.4  | 22.1 |
| 1M     | 26.2  | 22.8  | 19.4  | 15.2 | 15.8  | 16.6  | 20.4 |
| 3M     | 23.1  | 20.4  | 18.0  | 15.4 | 15.8  | 16.4  | 19.2 |
| 6M     | 21.4  | 19.2  | 17.4  | 15.6 | 15.8  | 16.2  | 18.4 |

Red intensity = put skew premium (OTM puts systematically more expensive) · Downward slope across expiries = normal term structure (contango) · ATM column always lowest vol per expiry

IMPLIED VOLATILITY SMIRK — PUT SKEW ELEVATED · BSM FLAT LINE (DASHED) vs ACTUAL (BLUE)  
Strike Offset from Spot | Y-axis: Implied Volatility %



### PUT SKEW / VOLATILITY SMIRK

Equity markets persistently trade with downside skew — OTM puts are systematically more expensive (higher IV) than equidistant OTM calls. This 'volatility smirk' exists because: (1) institutional investors consistently buy downside protection, (2) dealers charge more for the gamma risk in short puts, (3) crash risk has fat tails. The steeper the skew, the more the market fears a left-tail event.

### TERM STRUCTURE: CONTANGO vs BACKWARDATION

Normal (Contango): short-dated IV < long-dated IV — the market expects more uncertainty over time. Inverted (Backwardation): near-term IV > long-dated IV — signals acute near-term fear, often seen before FOMC, earnings, or CPI events. Backwardation is a key stress signal and creates calendar spread opportunities for informed traders.

**IV RANK (IVR) & IV PERCENTILE (IVP)**

$IVR = (\text{Current IV} - 52W \text{ Low}) / (52W \text{ High} - 52W \text{ Low})$ .  $IVP = \%$  of days in past year with lower IV. High IVR/IVP means options are expensive vs history — favor selling premium. Low IVR/IVP means options are cheap — favor buying optionality. Example: IVR of 24% means vol is in the lower quarter of its annual range — cheap to own options here.

**VVIX (VOL OF VOL)**

VVIX measures the implied volatility of VIX options — essentially the 'volatility of volatility.' Elevated VVIX (above 100) signals that vol itself is expected to be unstable, compressing the edge for volatility sellers and increasing the cost of tail hedges. Low VVIX = stable vol environment.

05

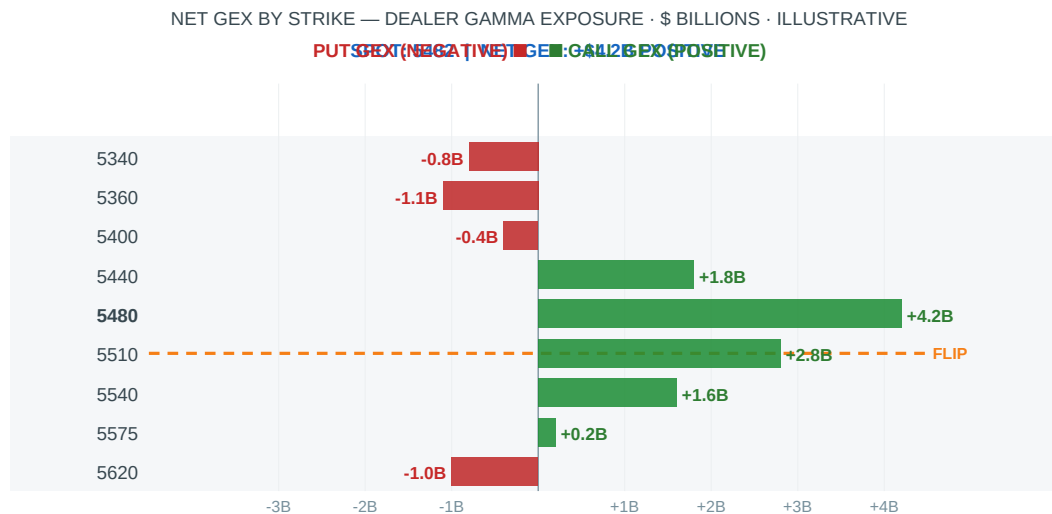
## GEX · DEX · DEALER HEDGING

GAMMA EXPOSURE · DELTA EXPOSURE · MECHANICAL FLOWS

**GEX (Gamma Exposure)** and **DEX (Delta Exposure)** estimate the aggregate hedging obligations of options market makers. Because dealers typically take the opposite side of customer flow — when customers buy calls, dealers are short calls and must buy the underlying to delta-hedge — these hedging flows create **systematic, non-fundamental price effects**. This mechanical structure is the core of options flow analysis. The size and direction of GEX at each strike determines whether dealer hedging *stabilizes* or *amplifies* price movement.

**IMPORTANT — DO NOT OVER-FIT DEALER HEDGE:** GEX levels identify zones of mechanical dealer activity, not guaranteed price walls or magnets. In low-liquidity conditions, thin OI, or when dealers have hedged via other instruments, specific GEX strikes may have little effect on realized price action. Always confirm with price action, the IV surface shape, and flow data before assigning high confidence to a specific GEX level as support or resistance. A single strike with large OI is not automatically a 'pin' without corroborating structure from multiple data layers.

## NET GEX BY STRIKE · SPX ILLUSTRATIVE · \$ BILLIONS



**POSITIVE GEX REGIME**  
(Dealers Long Gamma)

Dealers are net long gamma. As spot **RISES** toward a call-heavy strike, their delta increases so they **SELL** the underlying to stay neutral. As spot **FALLS**, they **BUY** the underlying. This counter-trend behavior acts as a structural stabilizer — compressing realized volatility, supporting dips, and capping rallies. Markets often range-bound near large positive GEX concentrations. Dip buyers structurally supported by dealer flows.

**NEGATIVE GEX REGIME**  
(Dealers Short Gamma)

Dealers are net short gamma. As spot **FALLS**, their delta becomes more negative, so they must **SELL** the underlying to re-hedge. As spot **RISES**, they **BUY**. This pro-trend behavior **AMPLIFIES** moves — down days extend further, up days have more momentum. Negative GEX environments correlate with elevated realized volatility, gap risk, and momentum continuation. This is when the structural stabilizer disappears from the market.

**GEX FLIP ZONE**

The price level where net dealer gamma transitions from positive to negative. Below this line, the stabilizing dealer bid disappears and flows become momentum-accelerating. Identifying the flip zone provides context for when a sell-off can self-reinforce versus when structural support is likely to hold. In the illustrative example: GEX flip near 5,440 — a sustained break below that level would shift the regime toward negative gamma dynamics.

**DEX (DELTA EXPOSURE)**

DEX estimates the net delta that dealers carry across all open options positions. Negative DEX means dealers are net short delta overall and must be long the underlying to stay hedged — this creates a structural bid in the futures/cash market. Positive DEX means dealers are net long delta and will hedge by selling the underlying. DEX direction can explain why certain rallies or sell-offs find unexpected buyers or sellers.

06

OPTIONS PRICING

BLACK-SCHOLES · INPUTS · PREMIUM DECOMPOSITION · BREAK-EVEN

The **Black-Scholes-Merton (BSM)** model provides the foundational framework for option pricing — a closed-form solution for European-style options. While BSM has known limitations (constant volatility, log-normal returns, no jumps), it remains the universal language for quoting and comparing options via **implied volatility**. Every professional options price is quoted in IV terms precisely because it converts the dollar premium into a standardized vol unit, enabling apples-to-apples comparison across strikes and expiries.

BLACK-SCHOLES-MERTON FORMULA

| FORMULA                                                                                                                                                                                           | INPUT VARIABLES                                                                                                                                                                                                                                             | ASSUMPTIONS & LIMITATIONS                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $C = S \cdot N(d1) - K \cdot e^{(-rT)} \cdot N(d2)$ $P = K \cdot e^{(-rT)} \cdot N(-d2) - S \cdot N(-d1)$ $d1 = [\ln(S/K) + (r + s^2/2) \cdot T] / s \cdot \sqrt{T}$ $d2 = d1 - s \cdot \sqrt{T}$ | <b>S</b> = Current spot price<br><b>K</b> = Strike price<br><b>r</b> = Risk-free interest rate<br><b>T</b> = Time to expiry (years)<br><b>s</b> = Implied volatility<br><b>N()</b> = Cumulative normal dist.<br><b>C</b> = Call price, <b>P</b> = Put price | <b>Assumes:</b> Constant vol, no dividends, no jumps, log-normal returns, continuous trading.<br><b>Real markets violate all of these</b> — which is precisely why the IV surface is not flat, why skew exists, and why extended models (Heston stochastic vol, SABR, local vol) were developed. |

PREMIUM DECOMPOSITION & BREAK-EVEN ANALYSIS

| CONCEPT                 | FORMULA / EXAMPLE                                                       | WHAT IT MEANS                                                                                                             |
|-------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Option Premium          | <b>Premium = Intrinsic + Extrinsic</b>                                  | Total cost of the contract. For OTM options, 100% is extrinsic. For deep ITM, mostly intrinsic.                           |
| Intrinsic Value         | <b>Call:</b> $\max(S - K, 0) = \max(5482 - 5400, 0) = \$82$             | The immediate exercise value — how much money you'd make if exercised today. Only ITM options have intrinsic value.       |
| Extrinsic Value         | <b>Extrinsic = Premium - Intrinsic = \$83.40 - \$82 = \$1.40</b>        | The optionality premium — payment for time remaining and volatility. Decays to zero at expiry regardless of position.     |
| Call Break-Even         | <b>B/E = Strike + Premium = 5480 + 18.55 = 5,498.55</b>                 | Spot must exceed 5,498.55 at expiry for the long call buyer to profit. Premium paid is a sunk cost.                       |
| Put Break-Even          | <b>B/E = Strike - Premium = 5480 - 18.10 = 5,461.90</b>                 | Spot must be below 5,461.90 at expiry for the long put buyer to profit. Direction AND magnitude must be right.            |
| Expected Move (1-sigma) | <b>EM = S x IV x sqrt(T/365) = 5482 x 0.158 x sqrt(30/365) = ±\$248</b> | Market's 1-standard-deviation expected price range (~68% probability). This is exactly what ATM straddle pricing implies. |

07

# OPTIONS DATA CONFLUENCE

LAYERING SIGNALS · STRUCTURE · FLOW · VOLATILITY

Professional options analysis never relies on a single metric. Each data point is one layer — **confluence occurs when multiple independent signals align** at the same price level or in the same direction. GEX alone is noise. GEX + OI concentration + IV term structure inversion + unusual flow converging on the same level = high-conviction structure. The framework below maps how these layers interact in a real analysis workflow.

CONFLUENCE LAYER MAP · SPX 5480 ZONE · ILLUSTRATIVE

| LAYER       | METRIC         | READING       | SIGNAL    | INTERPRETATION                                                                          |
|-------------|----------------|---------------|-----------|-----------------------------------------------------------------------------------------|
| STRUCTURE   | Net GEX        | +\$4.2B       | POSITIVE  | Dealers long gamma — stabilizing, mean-reverting mechanical flow in market              |
| STRUCTURE   | GEX Flip Zone  | 5,440         | WATCH     | Below 5440 → regime shifts negative → vol amplification risk increases                  |
| POSITIONING | Max Call OI    | 5,620 (62.8k) | PIN RISK  | Heavy call OI above spot — gravitational effect toward expiry via dealer gamma hedging  |
| POSITIONING | Max Put OI     | 5,400 (28.9k) | SUPPORT   | Large put OI below spot — dealers long gamma here, creates mechanical buying cushion    |
| VOLATILITY  | IV Rank (IVR)  | 24% — LOW     | CHEAP VOL | Options historically inexpensive vs prior year — favors buying optionality, not selling |
| VOLATILITY  | 25-Delta Skew  | +4.2 vol pts  | PUT PREM  | Market paying significant premium for downside — tail hedge demand elevated, skew steep |
| FLOW        | Put/Call Ratio | 0.92 — HIGH   | HEDGING   | Above-average put volume relative to calls — institutional hedging activity elevated    |
| FLOW        | Net DEX        | -\$1.1B       | NEG DELTA | Net negative delta exposure from options — dealers must be long underlying to offset    |
| TERM STRUCT | VIX Curve      | CONTANGO      | STABLE    | Near-term vol < long-term vol — no acute crisis event priced into near-dated expiries   |

UNUSUAL OPTIONS FLOW · LARGE PRINTS · ILLUSTRATIVE

| TIME  | EXPIRY | STRIKE | TYPE     | ACTION       | SIZE  | PREMIUM | SIGNAL        |
|-------|--------|--------|----------|--------------|-------|---------|---------------|
| 09:32 | 04/04  | 5500   | CALL     | BUY TO OPEN  | 2,400 | \$440K  | BULL SWEEP    |
| 09:58 | 04/17  | 5400   | PUT      | BUY TO OPEN  | 1,800 | \$620K  | HEDGE BUY     |
| 10:44 | 04/30  | 5480   | STRADDLE | BUY TO OPEN  | 900   | \$1.12M | VOL BUY       |
| 11:20 | 05/16  | 5300   | PUT      | BUY TO OPEN  | 3,100 | \$1.88M | TAIL HEDGE    |
| 13:05 | 04/11  | 5520   | CALL     | SELL TO OPEN | 4,200 | \$280K  | COVERED WRITE |

NET CONFLUENCE READ: Positive GEX provides structural stabilization near current spot. Elevated skew and above-average P/C ratio signal institutional hedging demand. Low IVR (24%) means options are cheap relative to history — asymmetric risk/reward favors owning protection rather than selling it here. No backwardation in the VIX curve means the market is not pricing an imminent acute crisis, but the tail hedge flow (5300 put buy, \$1.88M) suggests smart money is positioning for a left-tail event beyond the immediate horizon. No single layer is decisive — the full picture is required.

## 08

## CORE OPTION STRUCTURES

SPREADS · COMBINATIONS · RISK PROFILES

## LONG CALL

Directional · Debit · Bullish

Buy a call at chosen strike. Long delta, long gamma, short theta, long vega. Profits from upside move AND/OR IV expansion. Maximum loss is premium paid. Requires spot to move above break-even (strike + premium) by expiry. Sensitive to IV crush after events.

## LONG PUT

Directional · Debit · Bearish

Buy a put at chosen strike. Short delta, long gamma, short theta, long vega. Profits from downside move AND/OR IV expansion. Maximum loss is premium paid. Most commonly used as portfolio tail hedge. Put skew means OTM puts are expensive — often better value through put spreads.

## BULL CALL SPREAD

Directional · Debit · Defined Risk/Reward

Buy lower-strike call, sell higher-strike call. Long delta (net), reduced gamma, reduced theta cost vs naked long call. Selling the OTM call reduces premium cost and IV risk, but caps maximum profit at the width of the spread minus debit paid. Best when expecting moderate directional move, not explosive rally.

## IRON CONDOR

Neutral · Credit · Defined Risk

Sell OTM call spread + sell OTM put spread. Short delta (net zero at initiation), short gamma, long theta, short vega. Profits from time decay and low realized vol. Maximum profit = total credit received. Requires spot to stay inside short strikes through expiry. Benefits from IV crush post-event. Classic strategy in high-IV environments (IVR > 50).

## STRADDLE / STRANGLE

Volatility · Debit · Direction-Neutral

Straddle: buy ATM call + ATM put same strike/expiry. Strangle: buy OTM call + OTM put. Long gamma, long vega, short theta. Profits from large move in either direction OR IV expansion. Break-even requires realized vol to exceed implied vol over holding period. Strangles are cheaper but require larger moves. Often bought before known binary events when IV is low.

## COVERED CALL / CASH-SECURED PUT

Income · Credit · Yield Enhancement

Covered call: long underlying + sell OTM call. Collects premium, sacrifices upside above strike. Cash-secured put: hold cash + sell OTM put. Earns premium, obligation to buy underlying at strike if exercised. Both are short vega, long theta. Work best in high-IV environments where premium collected is substantial. Both reduce but do not eliminate directional risk.

## 09

## CORE PRINCIPLES

## REFERENCE SUMMARY · WHAT TO CARRY FORWARD

0  
1**IV IS THE PRICING LANGUAGE**

All option premiums reduce to an implied volatility number for standardized comparison. The IV surface — not the dollar price — is what skilled options traders read first.

0  
2**THE GREEKS ARE INTERCONNECTED**

Gamma and theta are always opposed: convexity is never free — you pay for it in daily time value erosion. Vega links to both — long options pay theta to hold gamma and vega exposure.

0  
3**SKEW ENCODES FEAR**

The steepness of put skew is the market's real-time fear gauge — it reflects institutional demand for downside protection and the collective view on left-tail probability.

0  
4**GEX IDENTIFIES MECHANICAL REGIMES**

Positive GEX = dealers stabilize price (buy dips, sell rips). Negative GEX = dealers amplify moves. The GEX flip zone is the threshold between these regimes. Neither is permanent.

0  
5**DO NOT OVER-FIT DEALER HEDGE**

GEX levels are probabilistic zones of elevated mechanical activity, not guaranteed price walls. Thin OI, alternate hedging instruments, or low liquidity can neutralize any level. Use GEX as one confluence layer, never as a standalone prediction.

0  
6**IVR / IVP DETERMINES STRATEGY BIAS**

When implied vol is cheap (low IVR) → favor buying optionality. When implied vol is expensive (high IVR) → favor selling premium. Context governs which regime produces edge.

0  
7**CONFLUENCE IS THE STANDARD**

No single metric — GEX, OI, skew, IVR, flow — is actionable alone. Professional analysis requires multiple independent signals to align before assigning conviction.

0  
8**REALIZED VS IMPLIED VOL IS THE EDGE**

The core options edge is identifying when realized volatility will deviate from implied vol. If  $RV > IV$  → own gamma. If  $RV < IV$  → sell theta. Structure, flow, and surface shape are tools for estimating which side of this bet is better priced.

All data, strikes, prices, Greeks, GEX figures, and flow examples in this document are entirely illustrative and for educational purposes only. This is not financial advice, investment advice, or a recommendation to buy or sell any security or derivative. Options trading involves significant risk of loss. Past positioning patterns do not guarantee future price behavior. Always conduct your own due diligence.